

# HARI VIGNESH RAO 23BD1A052Q

## Experiment

### Creating and Analyzing Disk Images Using dc3dd and Autopsy (Alternative to FTK Imager)

#### 1. Aim

To create a forensic disk image and analyze digital evidence using **dc3dd** and **Autopsy** in **Kali Linux**.

#### 2. Objectives

After completing this experiment, students will be able to:

- Understand the concept of **digital forensics and disk imaging**
- Create a **bit-by-bit forensic copy of a storage device**
- Maintain **data integrity using hashing**
- Analyze disk images to identify **files, deleted files, and system artifacts**

#### 3. Introduction / Theory

##### Digital Forensics

Digital forensics is the process of **identifying, collecting, preserving, analyzing, and presenting digital evidence** from computers or storage devices.

Digital evidence may include:

- Documents
- Emails
- Browser history
- Logs
- Deleted files

To preserve evidence, investigators create a **forensic disk image**.

##### Disk Imaging

Disk imaging is the process of creating an **exact replica of a storage device**.

Characteristics of forensic disk images:

- Bit-by-bit copy of storage
- Preserves deleted files

- Maintains data integrity
- Allows investigation without altering original evidence

## Hashing in Forensics

Hashing is used to verify the **integrity of digital evidence**.

Common hash algorithms:

**Algorithm Purpose**

MD5 evidence integrity

SHA1 verification

SHA256 stronger security

If the hash value of the original disk and the disk image match, the **evidence is considered intact**.

## 4. Tools Used

### Tool Purpose

|            |                                     |
|------------|-------------------------------------|
| Kali Linux | forensic analysis environment       |
| dc3dd      | disk image acquisition              |
| Autopsy    | forensic investigation and analysis |

## 5. System Requirements

### Component Specification

|           |                    |
|-----------|--------------------|
| Processor | Intel i3 or higher |
| RAM       | Minimum 4 GB       |
| Disk      | 20 GB free space   |
| Software  | Kali Linux VM      |

## 6. Algorithm

1. Start the Kali Linux environment.
2. Generate activity on the system to create forensic artifacts.
3. Identify the disk partition using system commands.
4. Create a forensic disk image using dc3dd.
5. Generate hash values for integrity verification.
6. Start the Autopsy forensic analysis tool.
7. Import the disk image into Autopsy.
8. Analyze files, logs, and deleted artifacts.
9. Extract relevant digital evidence.

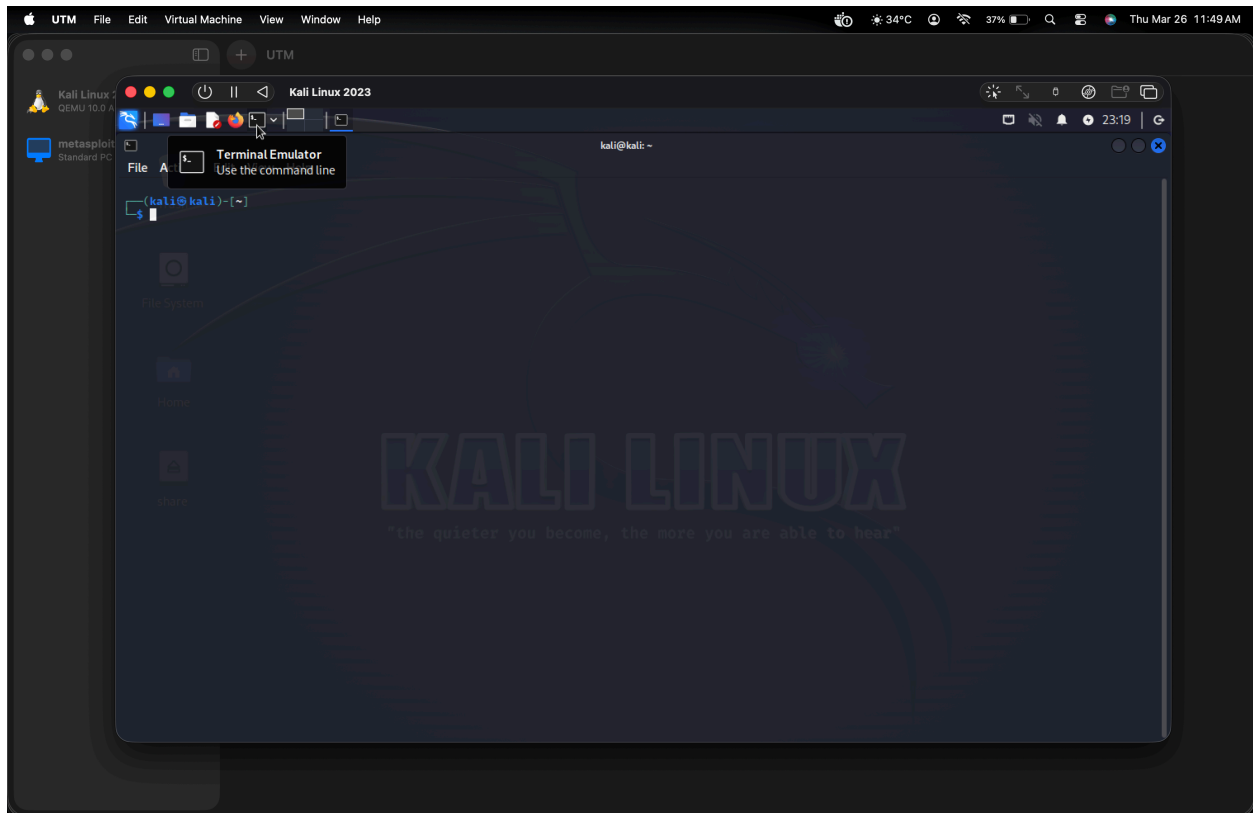
### 7. Procedure

#### Step 1: Open Terminal in Kali Linux

Press:

Ctrl + Alt + T

This opens the terminal.



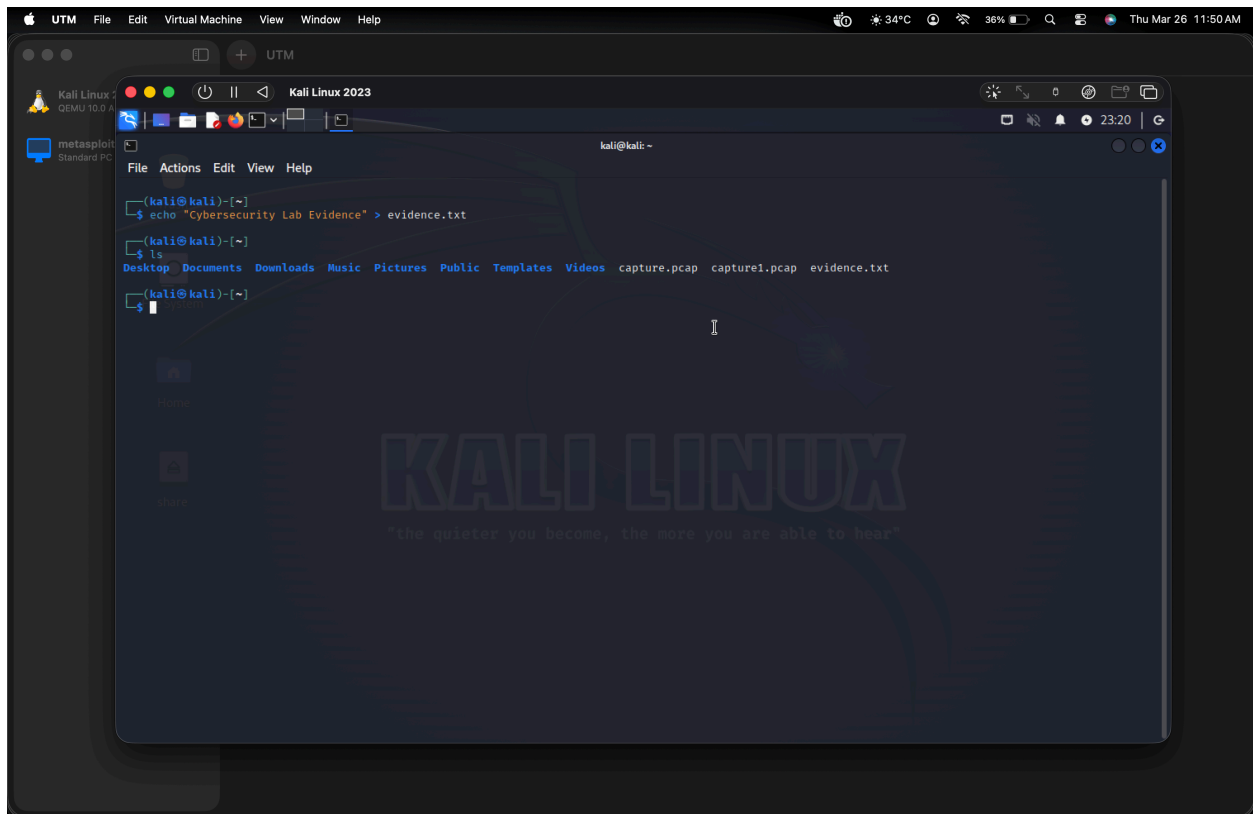
## Step 2: Create Sample Evidence

Generate a file that will later be discovered during forensic analysis.

```
echo "Cybersecurity Lab Evidence" > evidence.txt
```

Verify the file creation.

```
Ls
```



### Step 3: Identify Disk Partition

List storage devices.

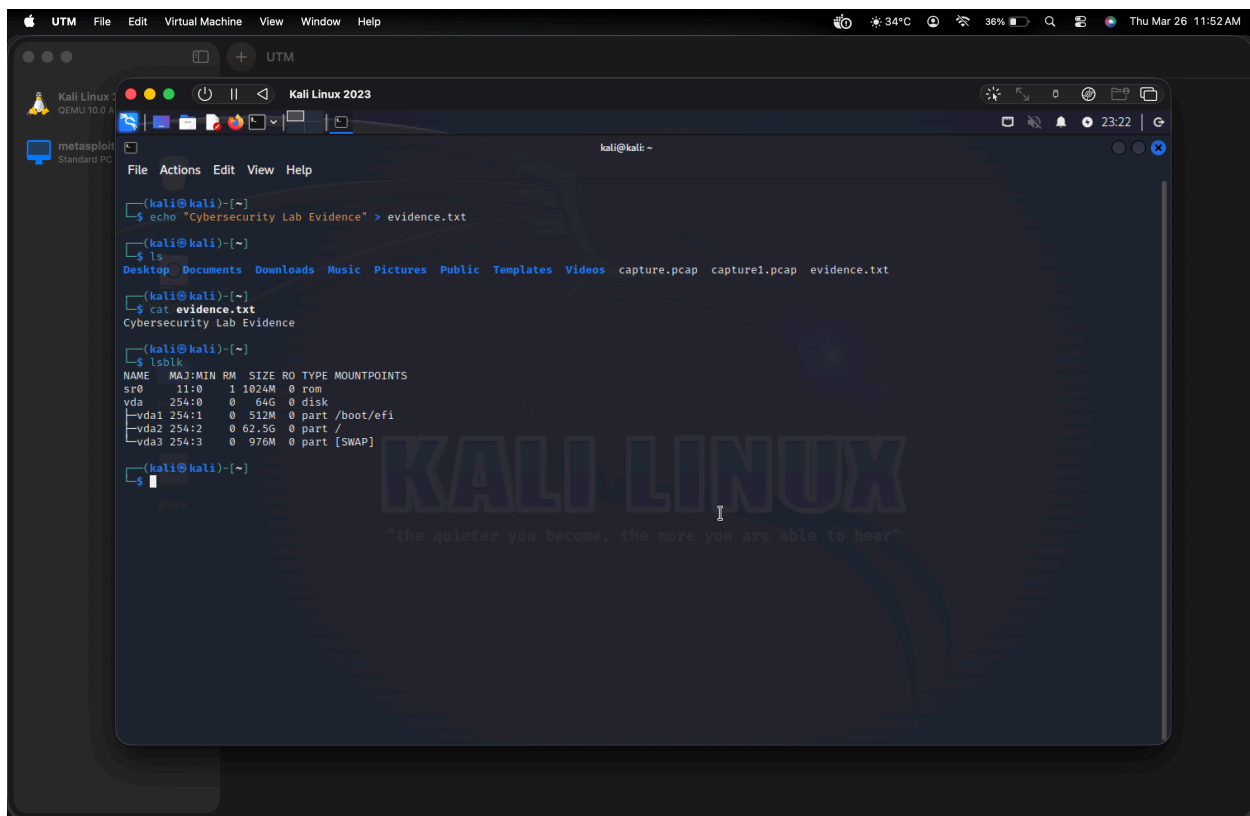
```
lsblk
```

Example output:

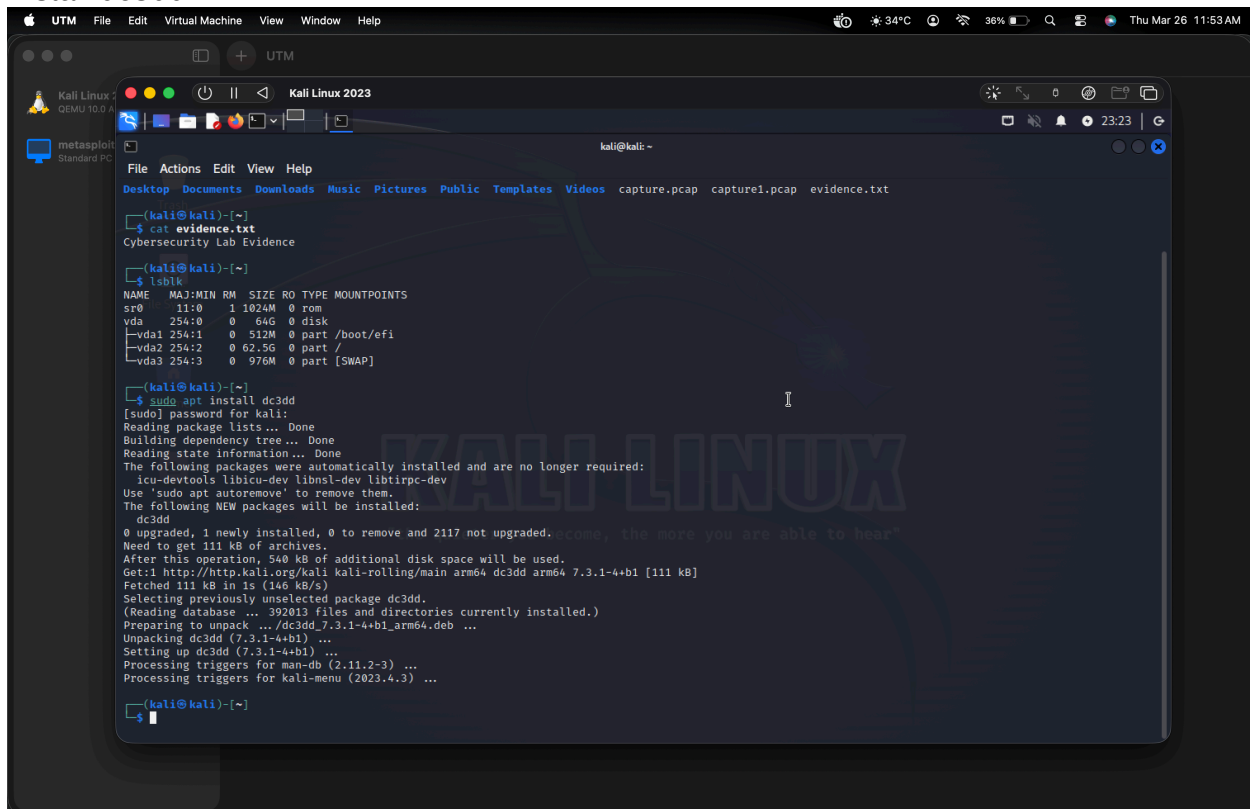
```
sda  
├─sda1  
└─sda2
```

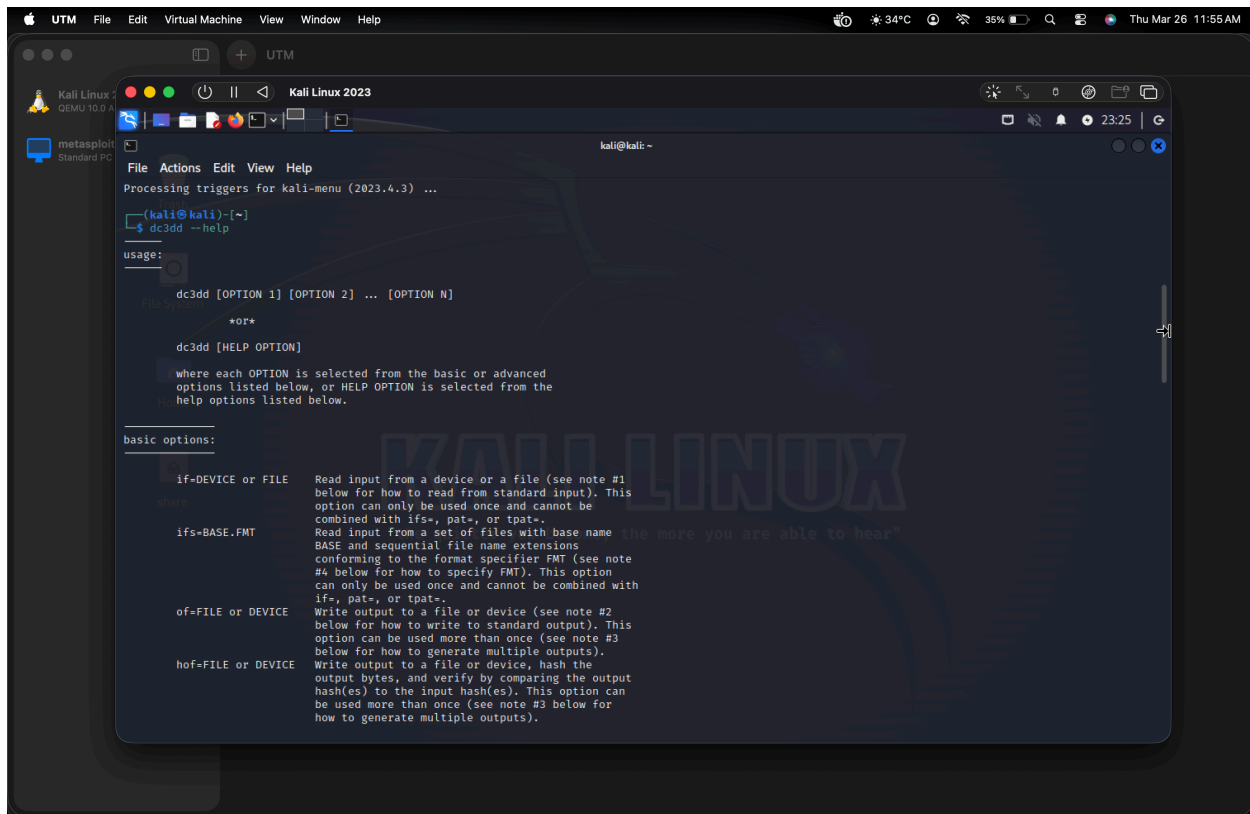
The partition **/dev/sda1** will be used for disk imaging





## Install dc3dd

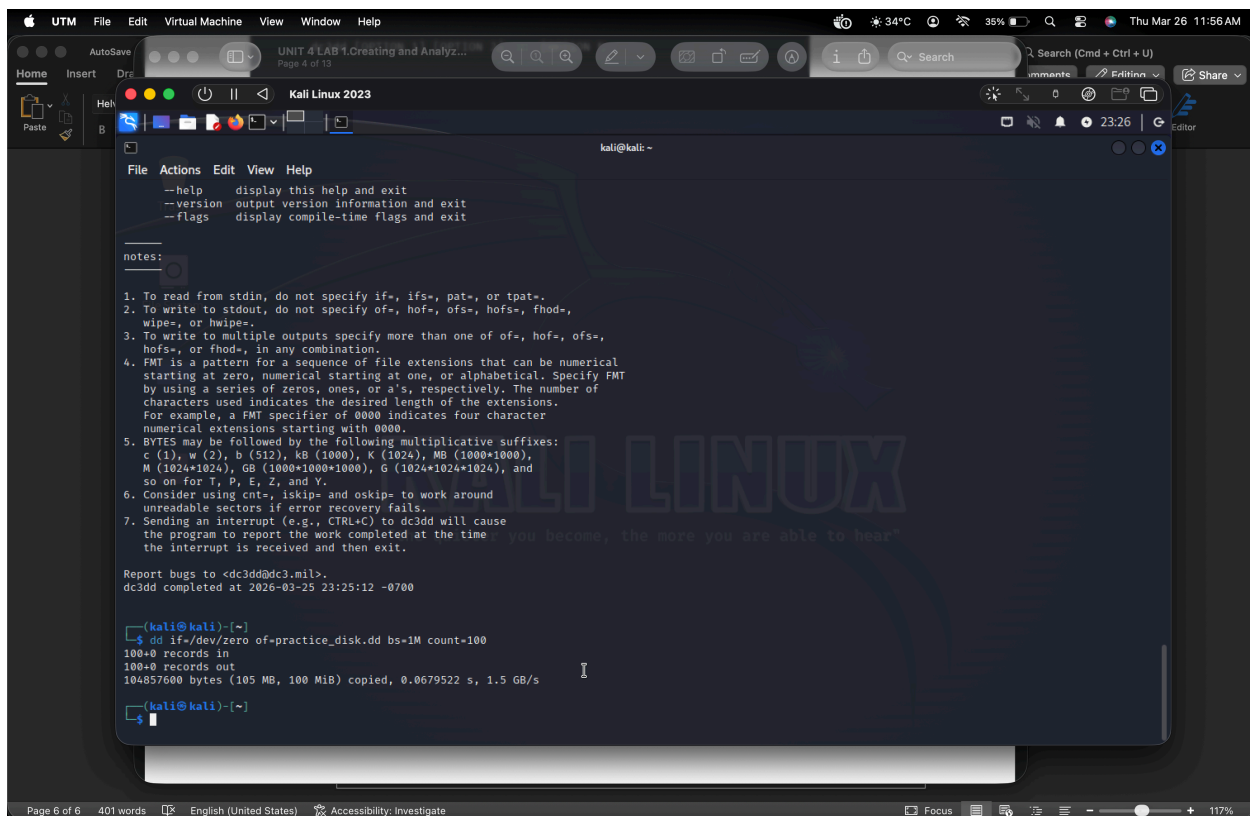




## Step 4: Create Disk Image

### Step 1: Create a Small Disk File

```
dd if=/dev/zero of=practice_disk.dd bs=1M count=100
```



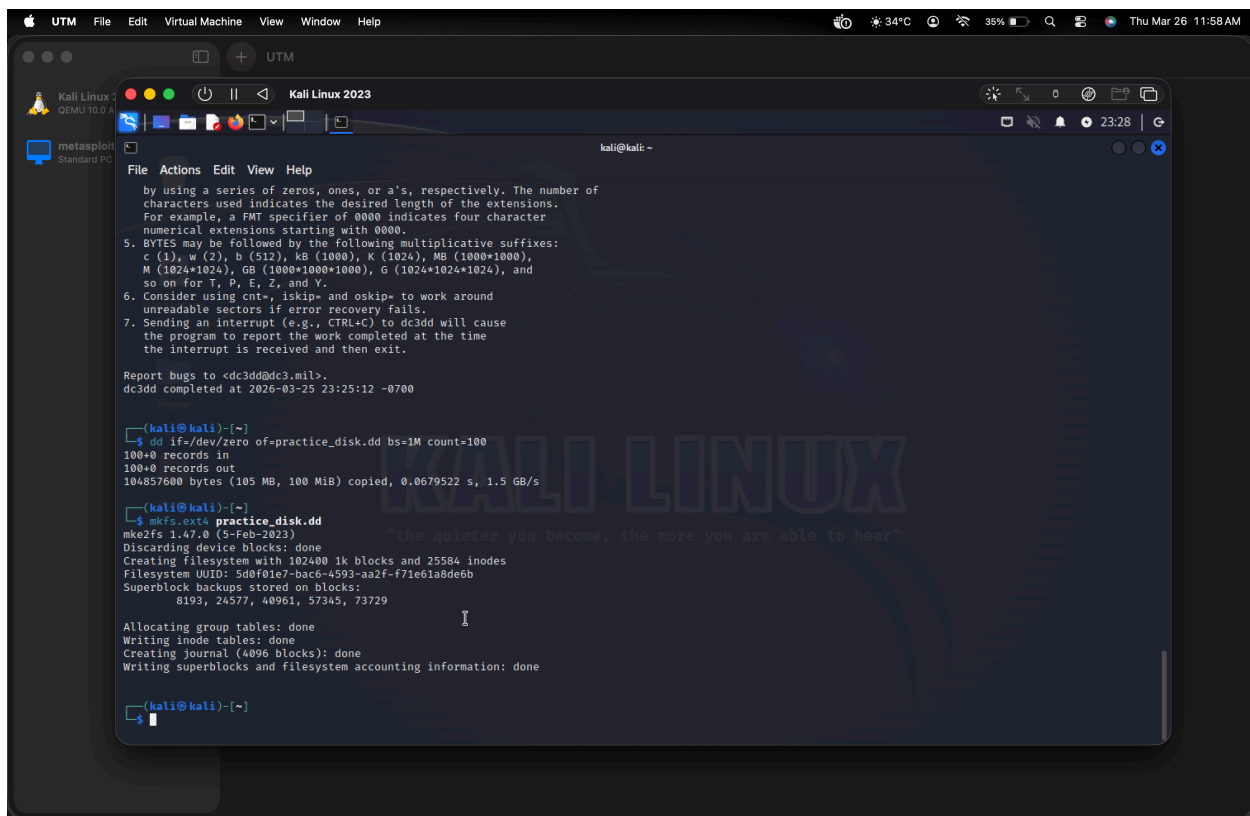
Explanation:

| Parameter | Description          |
|-----------|----------------------|
| if        | input disk           |
| of        | output image file    |
| hash      | generate MD5 hash    |
| log       | acquisition log file |

Creates a **100 MB** disk.

Step 2: Mount the Disk

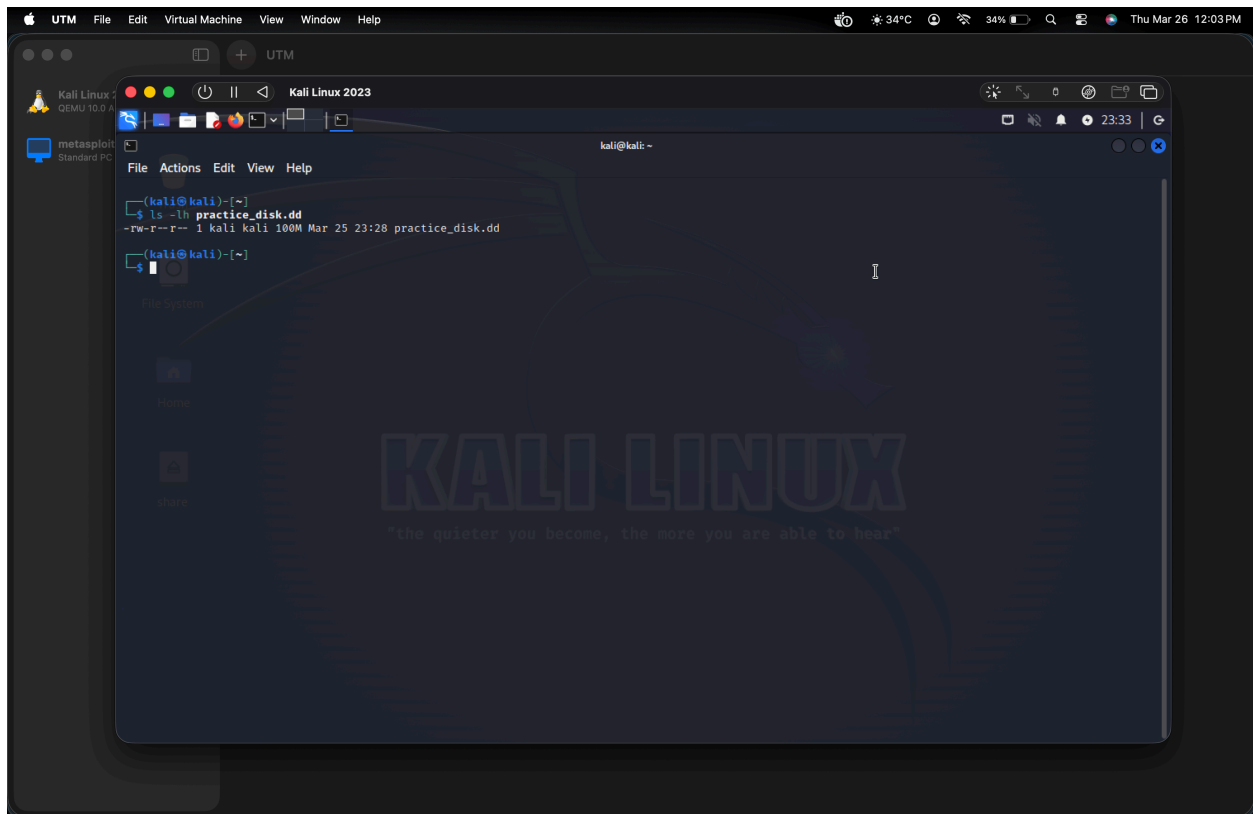
```
mkfs.ext4 practice_disk.dd
```



## Step 5: Verify Image Creation

Check the created image.

```
ls -lh practice_disk.dd
```



Check log file.

```
cat acquisition.log
```

The log will contain:

- image size
- hash value
- acquisition details

```
sudo dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5  
log=acquisition.log
```

The screenshot shows a Kali Linux 2023 terminal window. The user has executed the command `dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log`. The output shows that the command completed successfully, copying 104857600 bytes (100 M) at a speed of 491 M/s. The user then executed `cat acquisition.log`, which displays the command line and the results of the copy operation.

```
(kali@kali)-[~]
└─$ sudo dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log
dc3dd 7.2.646 started at 2026-03-25 23:36:45 -0700
compiled options:
command line dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log
sector size: 512 bytes (assumed)
104857600 bytes ( 100 M ) copied ( 100% ), 0 s, 491 M/s

input results for file 'practice_disk.dd':
204800 sectors in
c8527e18480ce322798d7fc76576992f (md5)

output results for file 'forensic_image.dd':
204800 sectors out

dc3dd completed at 2026-03-25 23:36:45 -0700

(kali@kali)-[~]
└─$ cat acquisition.log
dc3dd 7.2.646 started at 2026-03-25 23:35:08 -0700
compiled options:
command line dc3dd if=disk_image.dd of=forensic_image.dd hash=md5 log=acquisition.log
[!] opening 'disk_image.dd' for validation probe: No such file or directory
dc3dd aborted at 2026-03-25 23:35:08 -0700

dc3dd 7.2.646 started at 2026-03-25 23:36:45 -0700
compiled options:
command line dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log
sector size: 512 bytes (assumed)
104857600 bytes ( 100 M ) copied ( 100% ), 0.203828 s, 491 M/s

input results for file 'practice_disk.dd':
204800 sectors in
c8527e18480ce322798d7fc76576992f (md5)

output results for file 'forensic_image.dd':
204800 sectors out
```

The screenshot shows a Kali Linux 2023 terminal window. The user has executed the command `dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log`. The output shows that the command completed successfully, copying 104857600 bytes (100 M) at a speed of 491 M/s. The user then executed `cat acquisition.log`, which displays the command line and the results of the copy operation.

```
(kali@kali)-[~]
└─$ sudo dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log
dc3dd 7.2.646 started at 2026-03-25 23:36:45 -0700
compiled options:
command line dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log
sector size: 512 bytes (assumed)
104857600 bytes ( 100 M ) copied ( 100% ), 0 s, 491 M/s

input results for file 'practice_disk.dd':
204800 sectors in
c8527e18480ce322798d7fc76576992f (md5)

output results for file 'forensic_image.dd':
204800 sectors out

dc3dd completed at 2026-03-25 23:36:45 -0700

(kali@kali)-[~]
└─$ cat acquisition.log
dc3dd 7.2.646 started at 2026-03-25 23:35:08 -0700
compiled options:
command line dc3dd if=disk_image.dd of=forensic_image.dd hash=md5 log=acquisition.log
[!] opening 'disk_image.dd' for validation probe: No such file or directory
dc3dd aborted at 2026-03-25 23:35:08 -0700

dc3dd 7.2.646 started at 2026-03-25 23:36:45 -0700
compiled options:
command line dc3dd if=practice_disk.dd of=forensic_image.dd hash=md5 log=acquisition.log
sector size: 512 bytes (assumed)
104857600 bytes ( 100 M ) copied ( 100% ), 0.203828 s, 491 M/s

input results for file 'practice_disk.dd':
204800 sectors in
c8527e18480ce322798d7fc76576992f (md5)

output results for file 'forensic_image.dd':
204800 sectors out

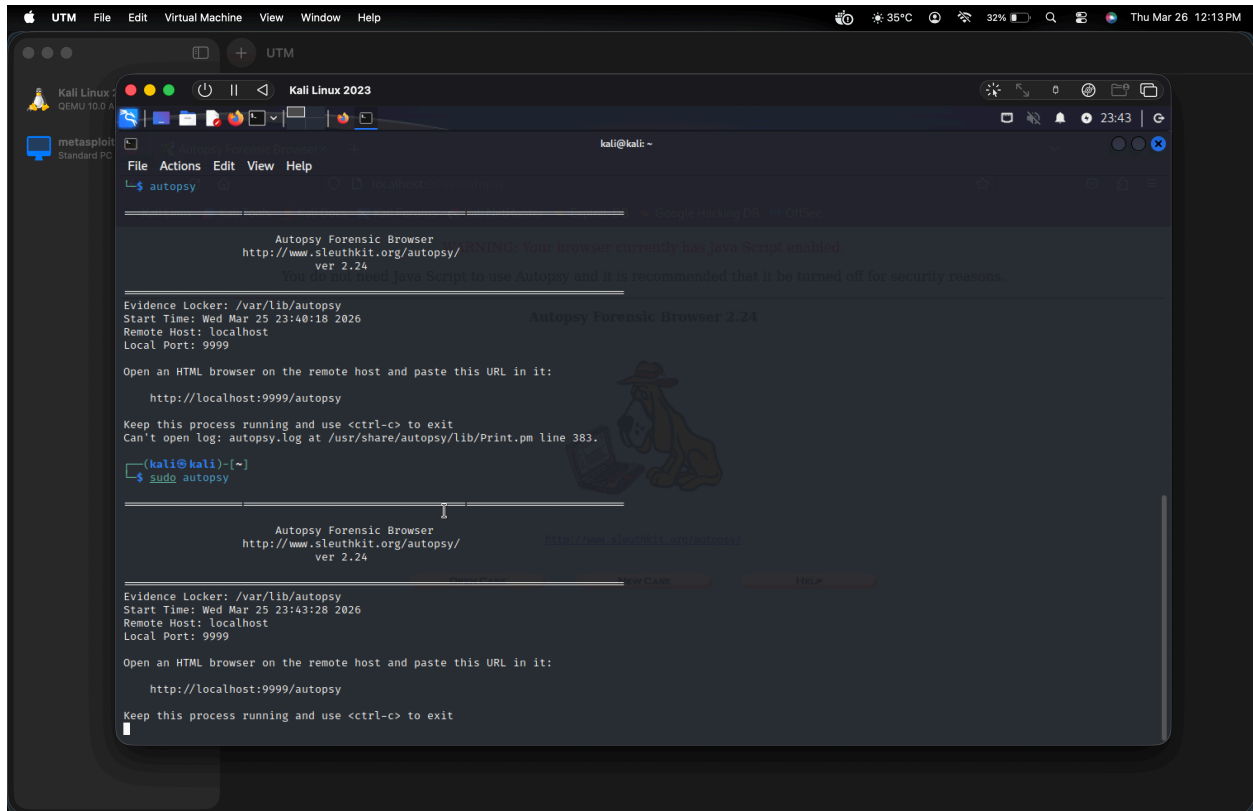
dc3dd completed at 2026-03-25 23:36:45 -0700

(kali@kali)-[~]
└─$
```

## Step 6: Start Autopsy

Launch the forensic analysis tool.

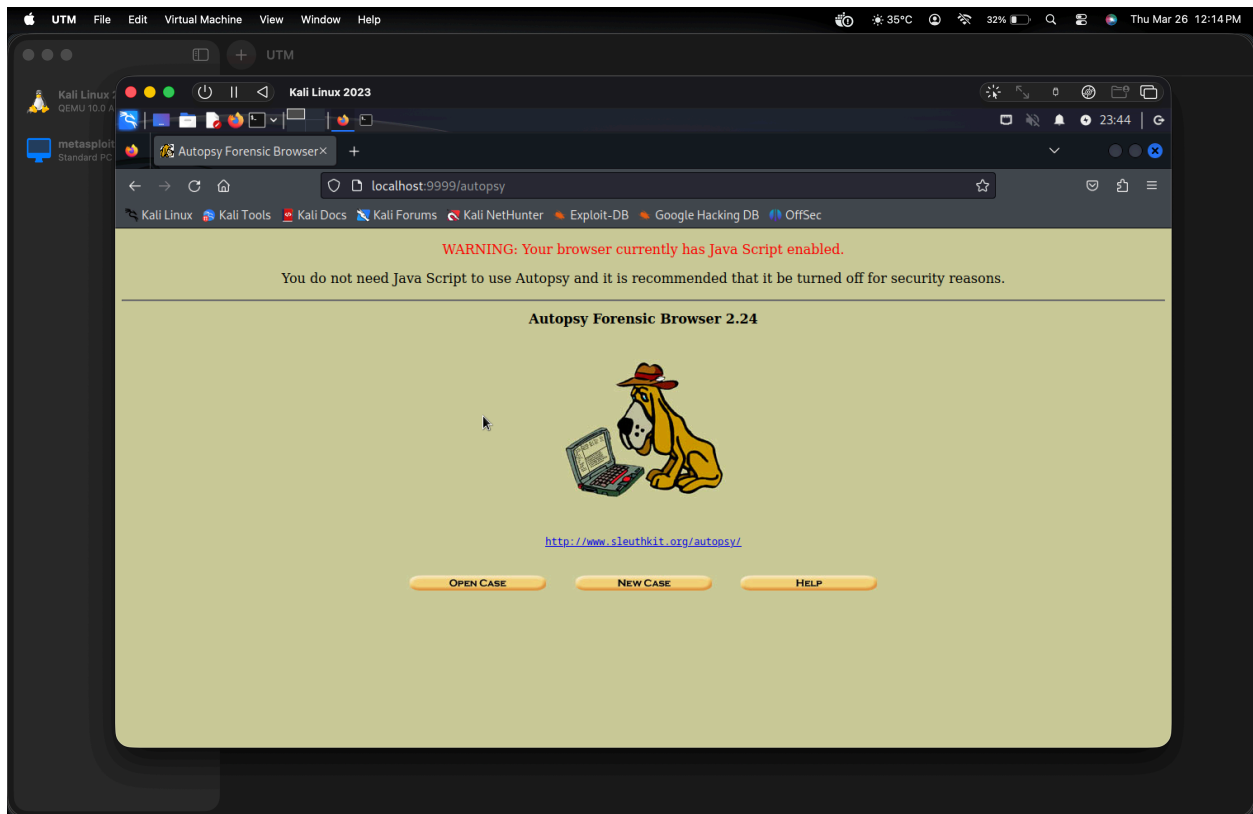
## Autopsy



The terminal will display a URL:

`http://localhost:9999/autopsy`

Open this link in the browser.

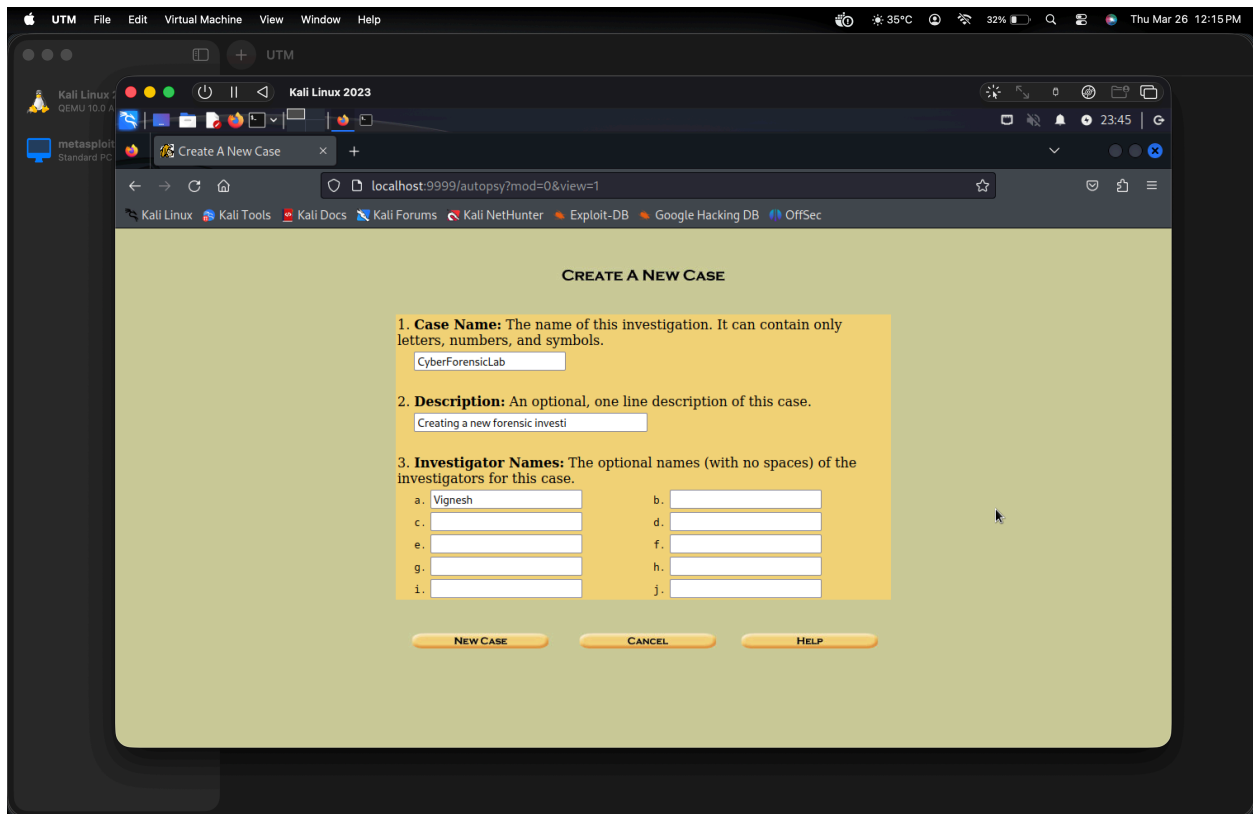


## Step 7: Create a New Case

In the Autopsy interface:

1. Click **Create New Case**
2. Enter case details:





## Step 8: Add Disk Image

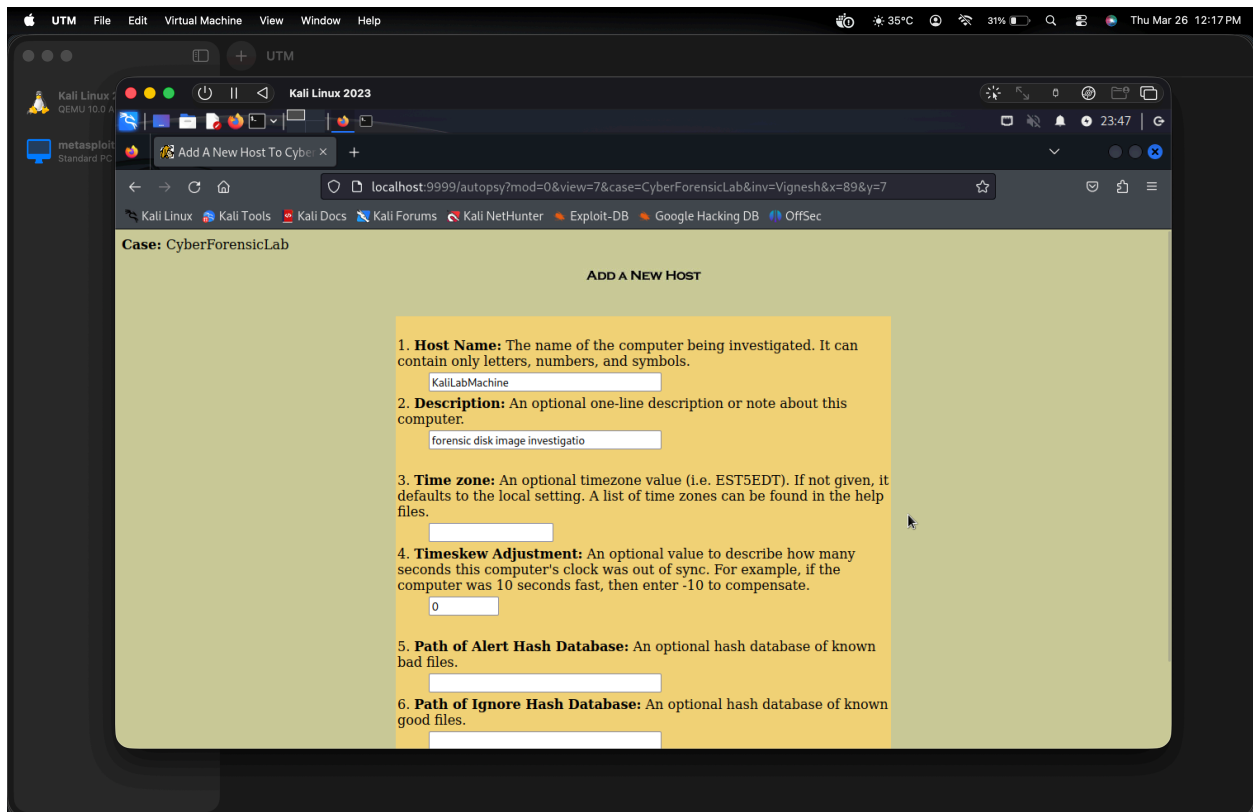
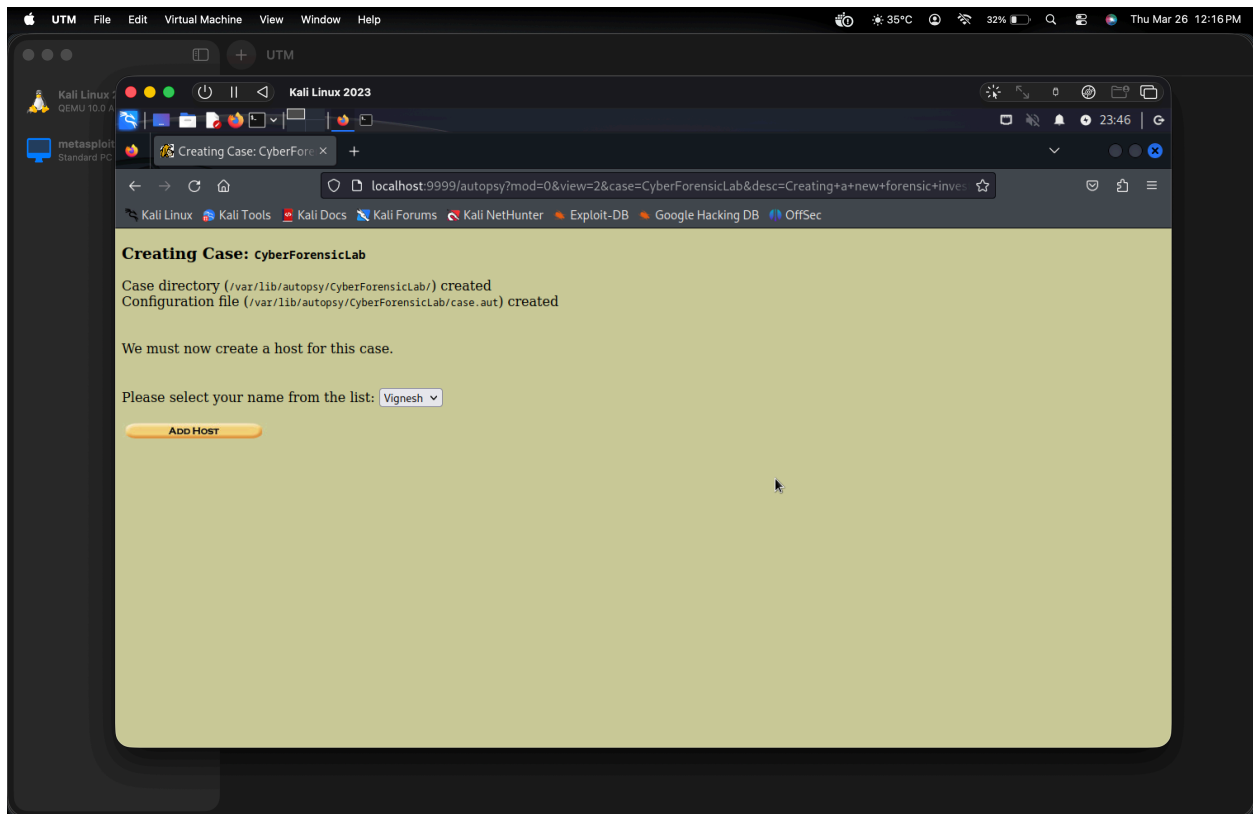
Select:

Add Host → Add Image

Browse and select:

`/home/kali/disk_image.dd`

Autopsy will begin analyzing the disk image.



## **Recommended Values**

### **Host Name**

KaliLabMachine

### **Description**

Kali Linux virtual machine used for forensic disk image investigation

### **Time Zone**

Leave Blank

### **Time Skew Adjustment**

Leave Blank

### **Alert Hash Database**

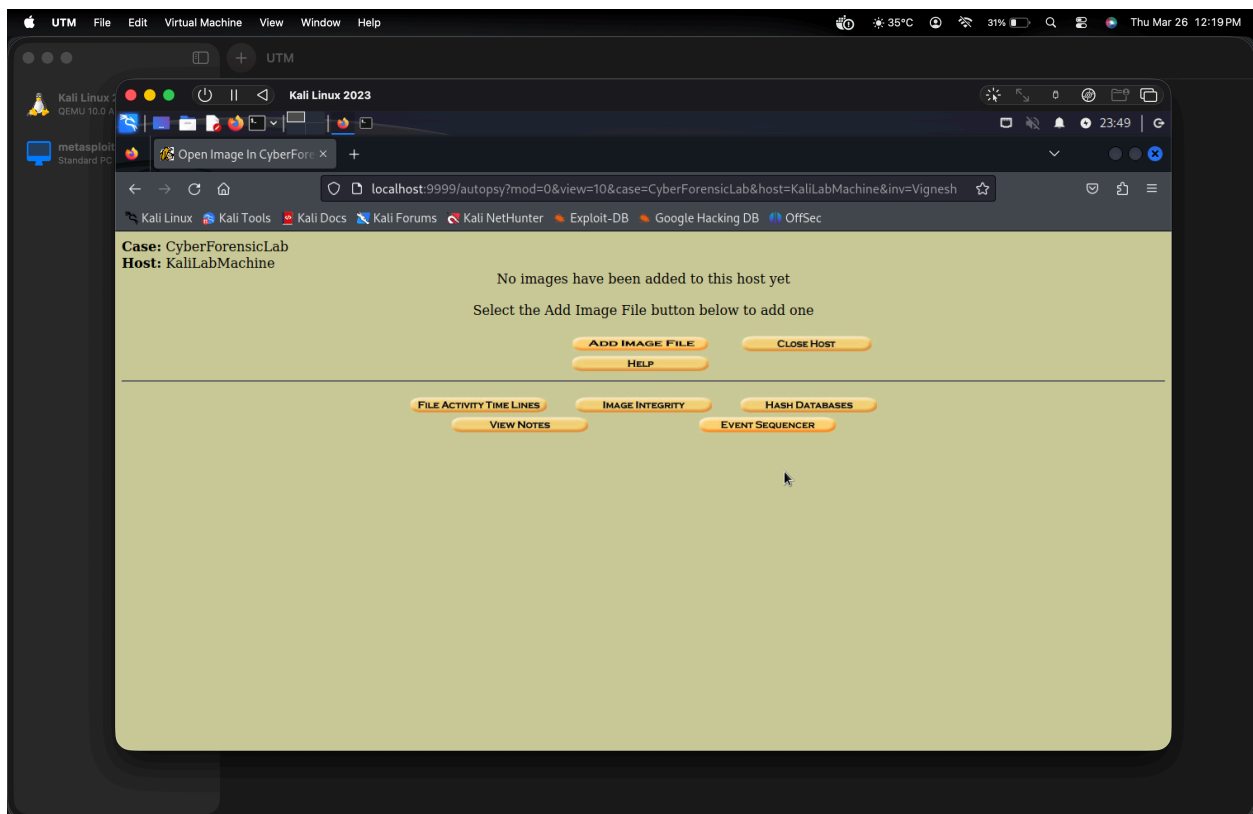
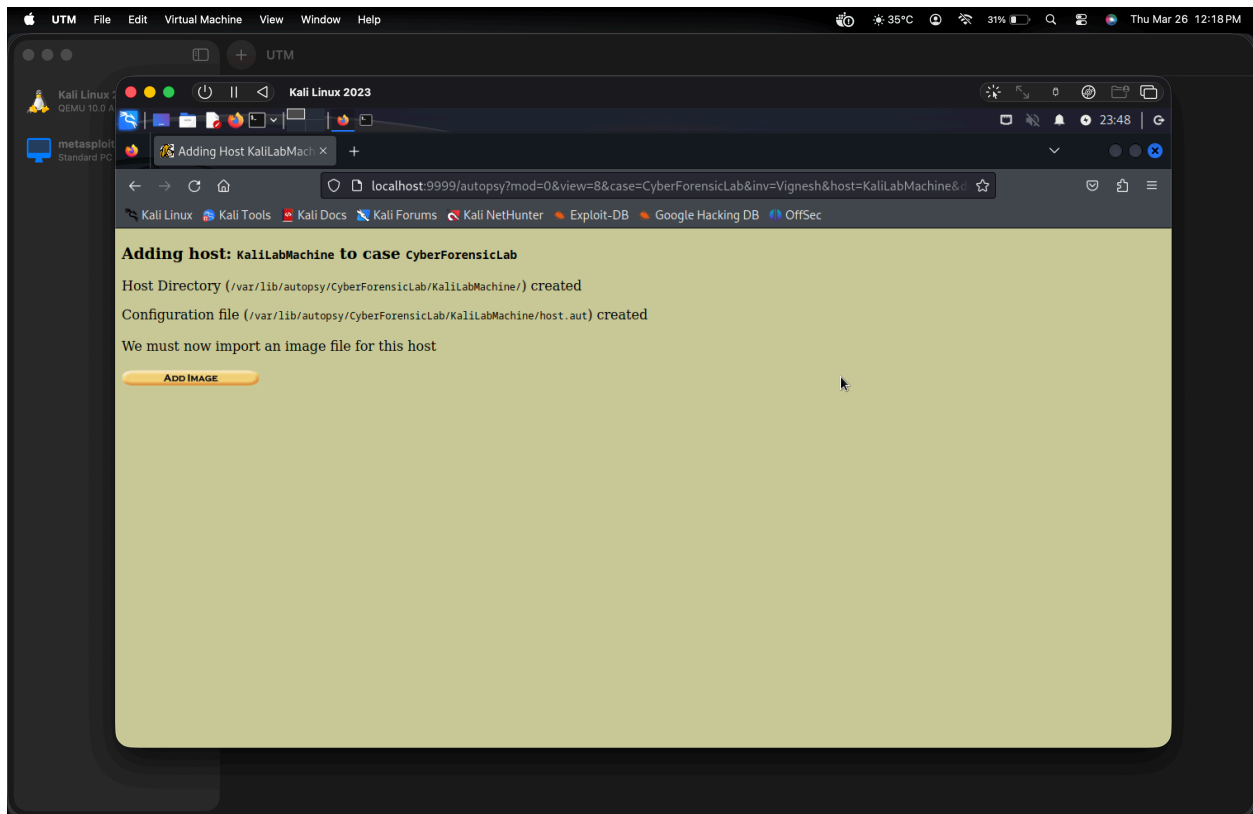
Leave Blank

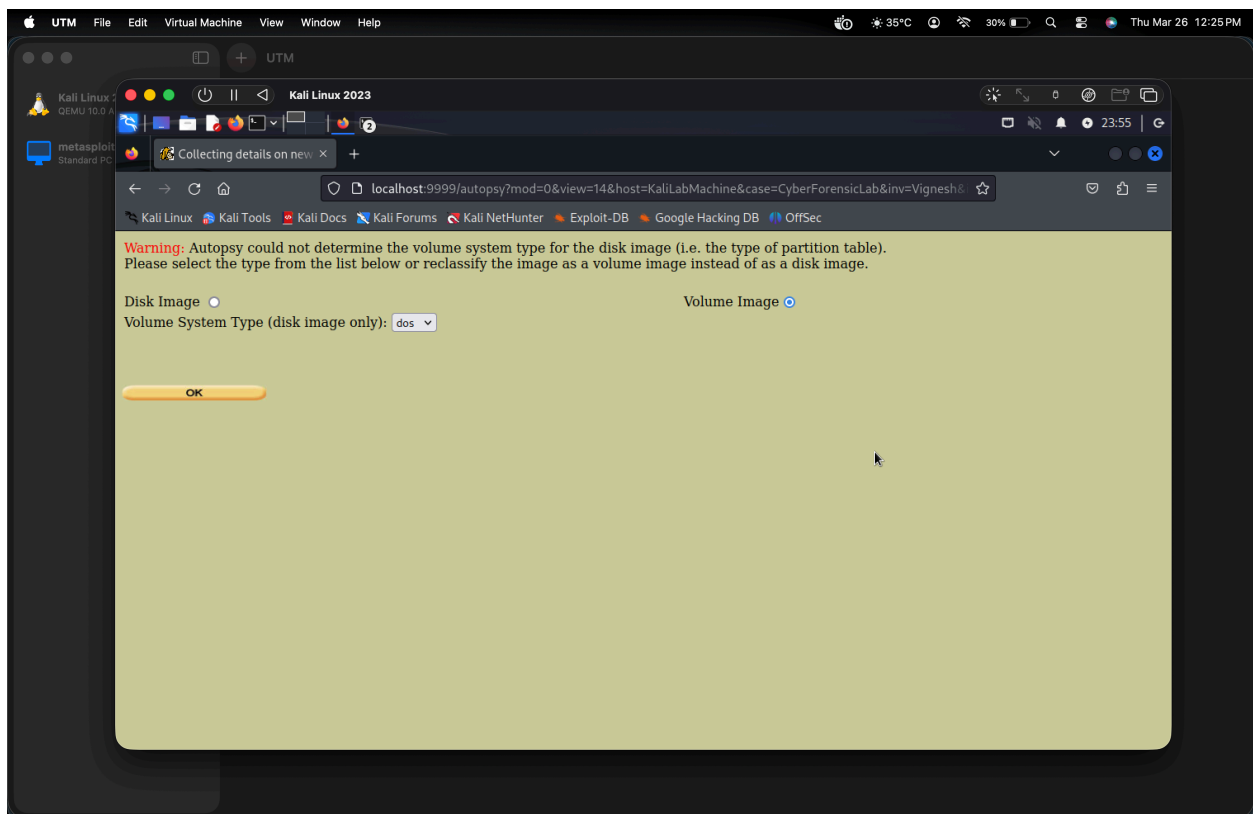
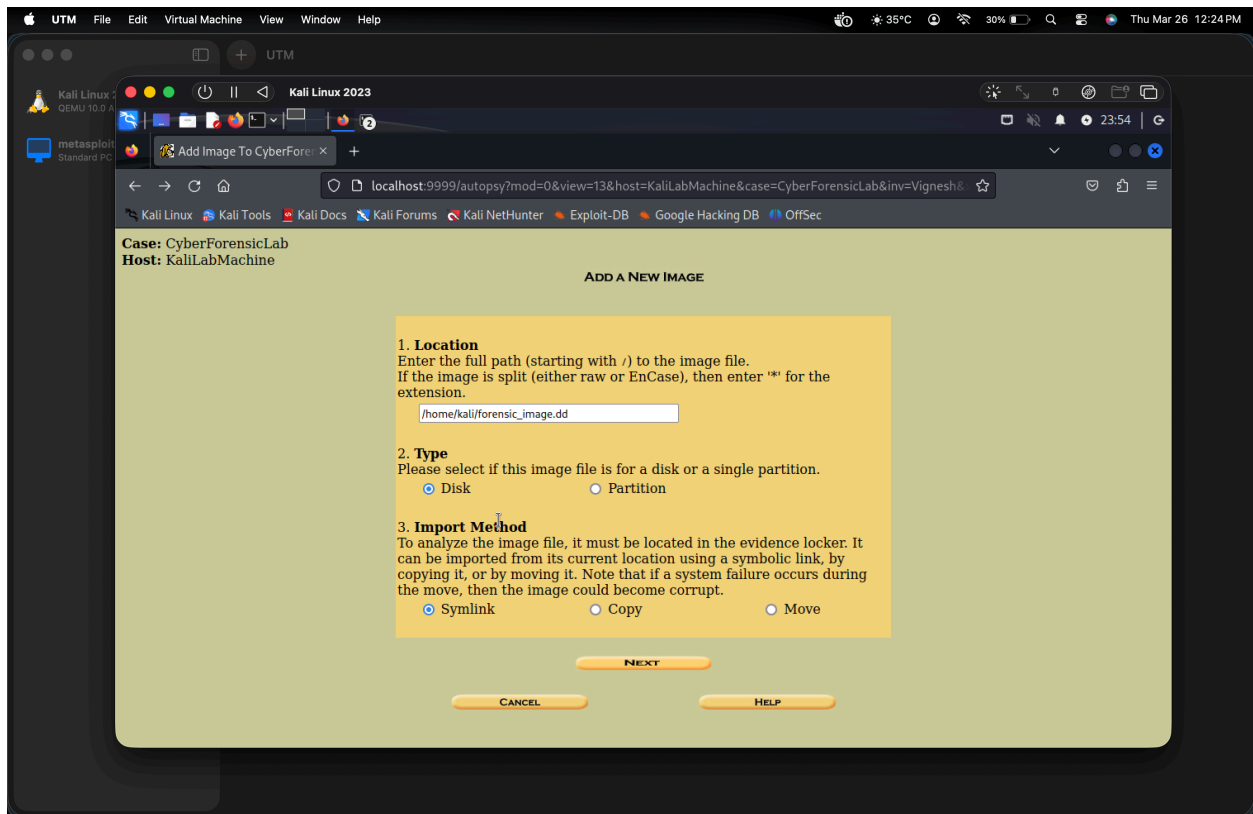
### **Ignore Hash Database**

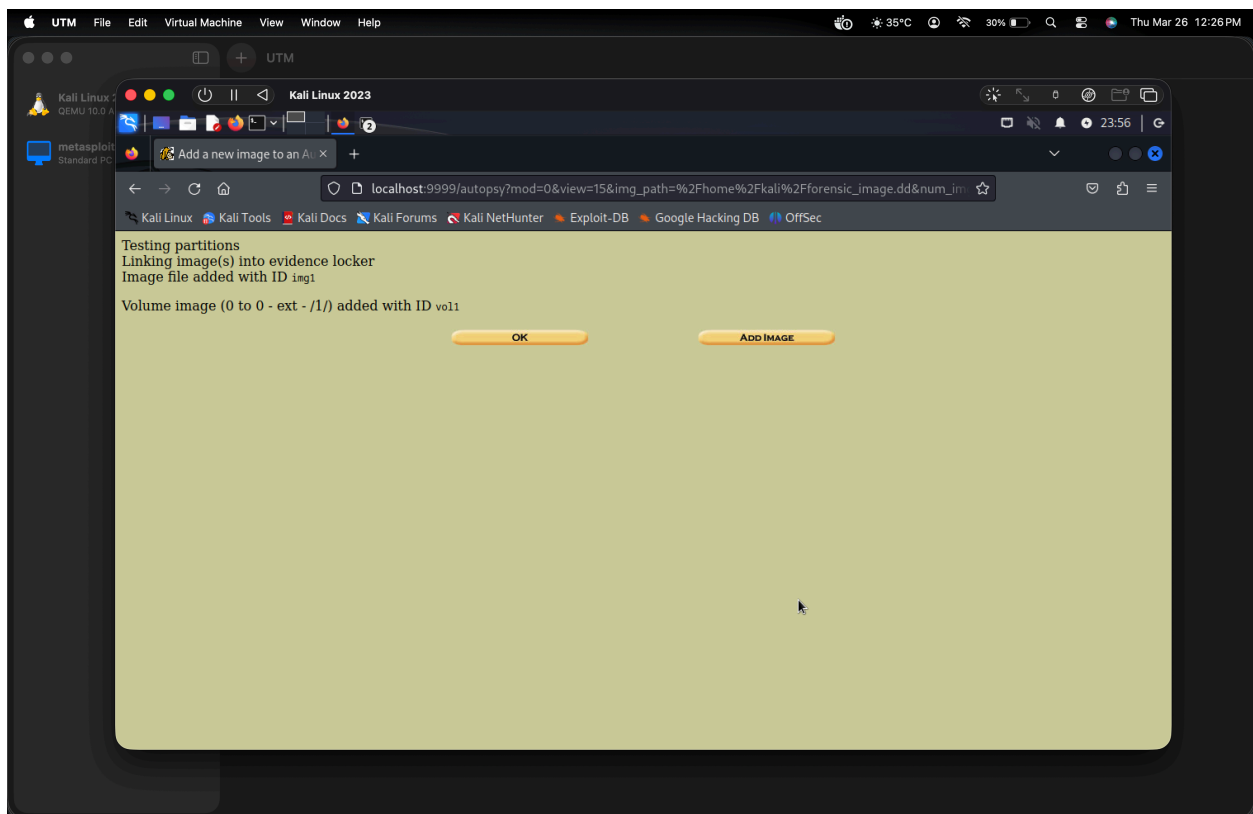
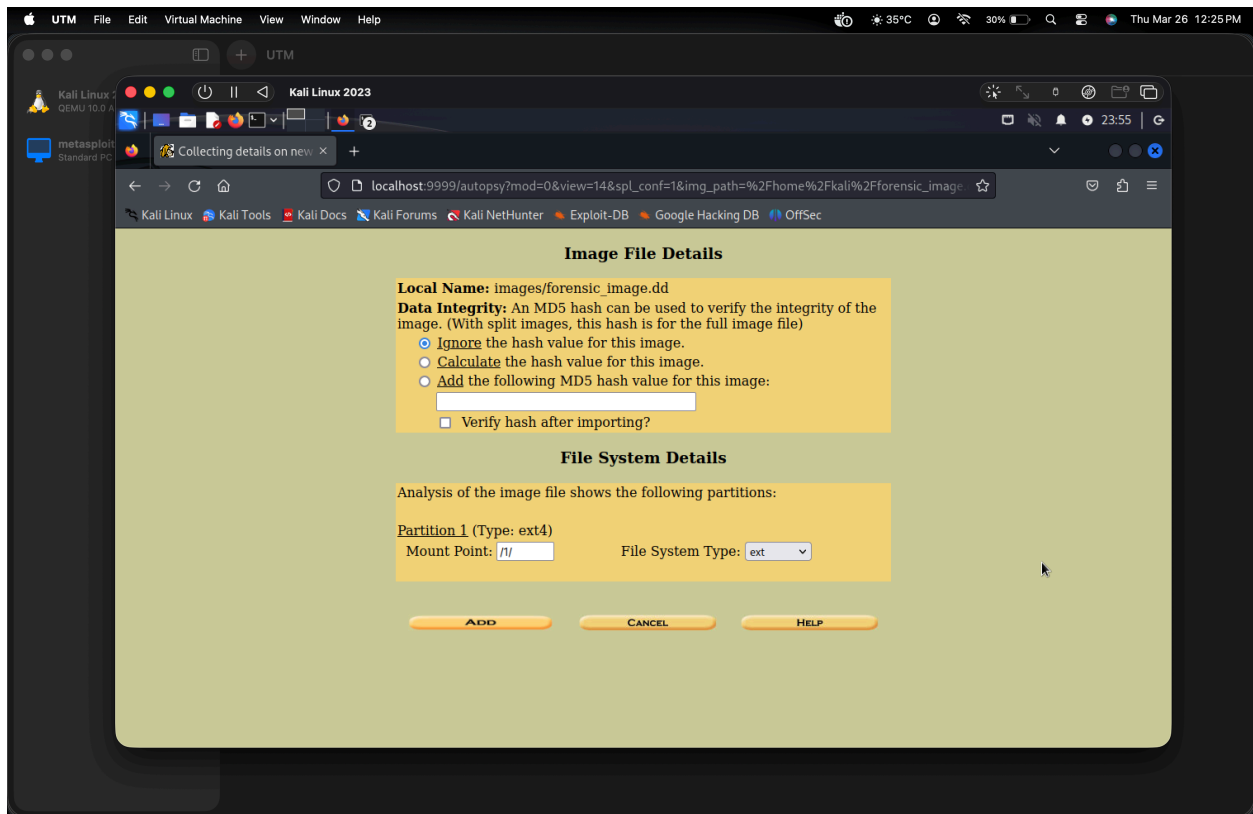
Leave Blank

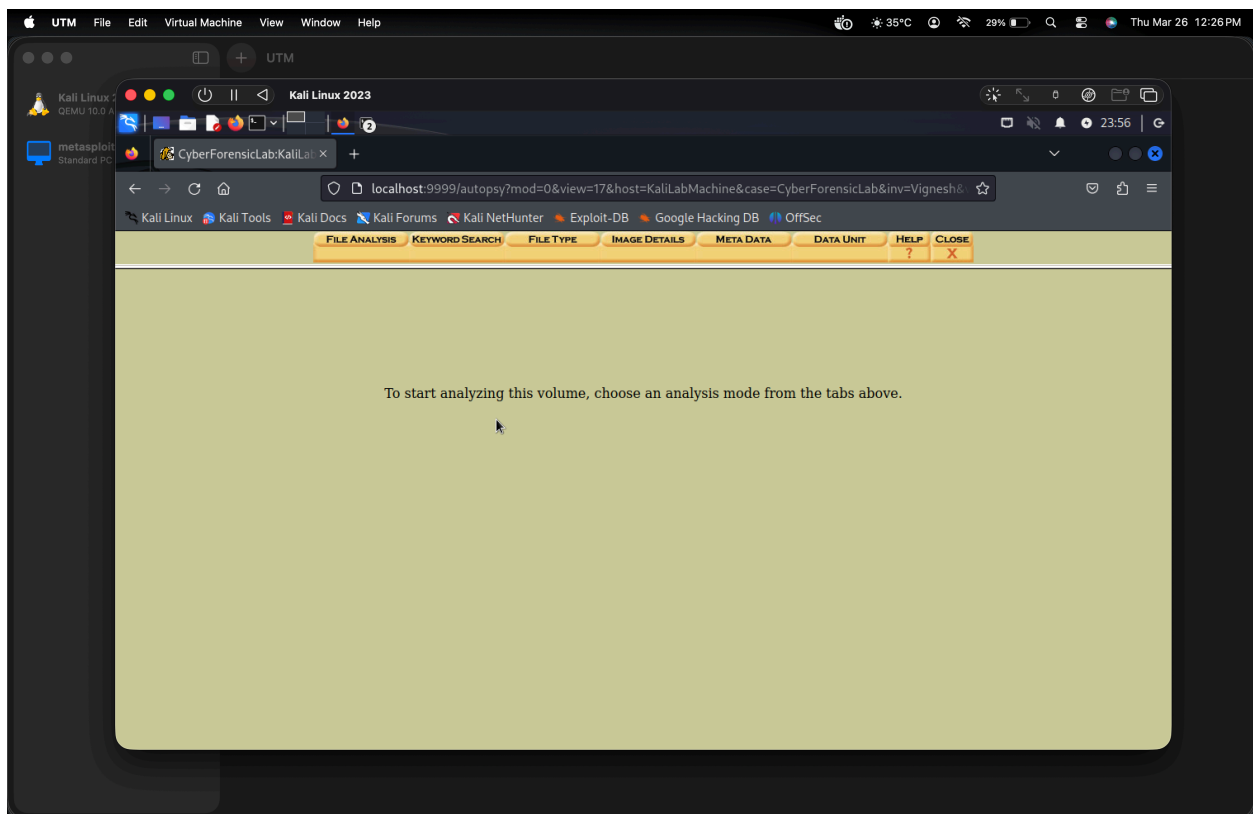
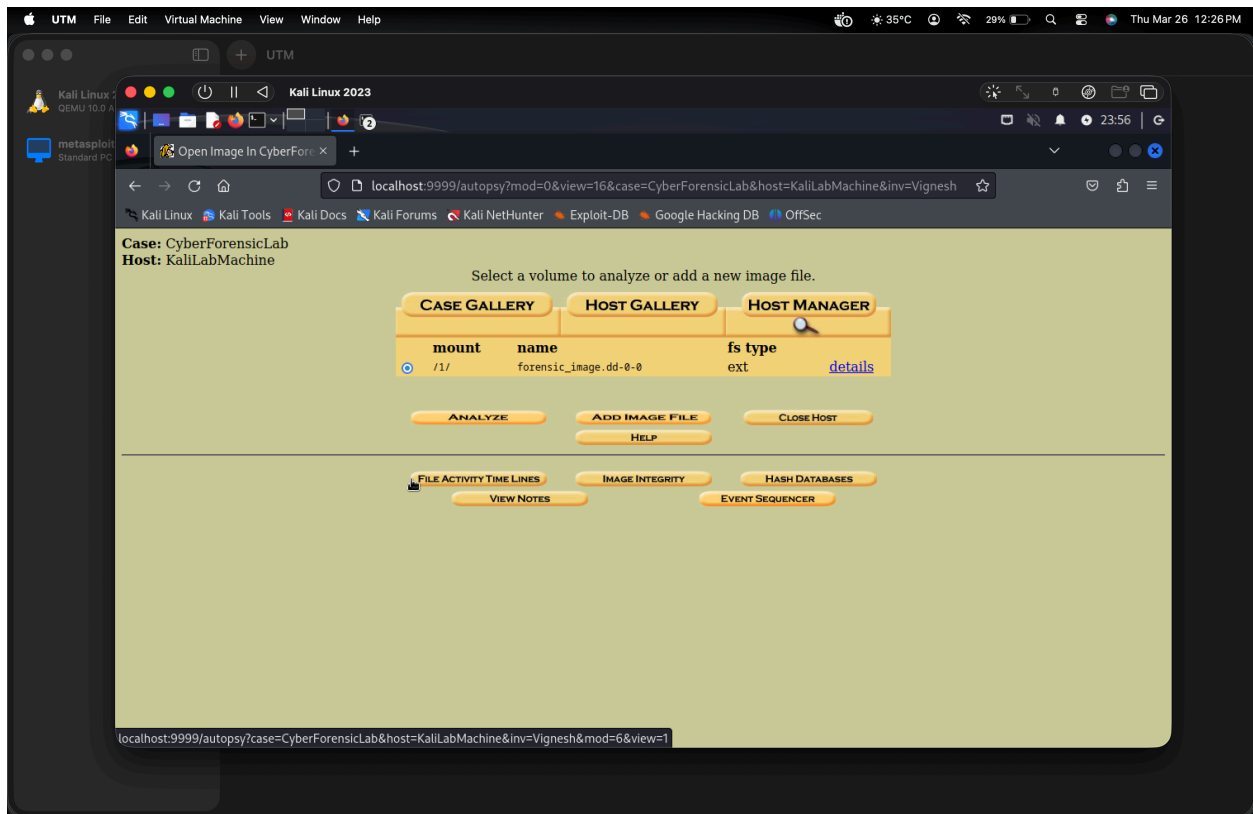
Then click:

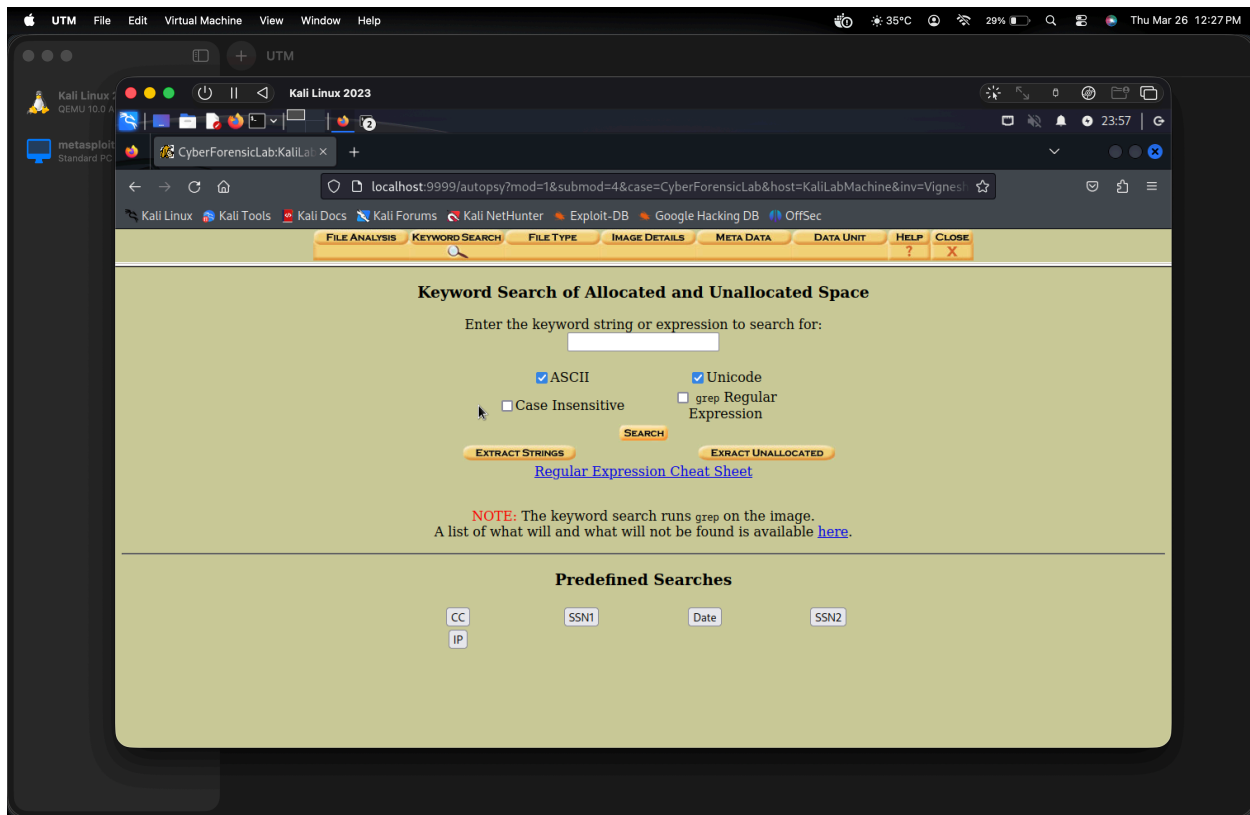
Next



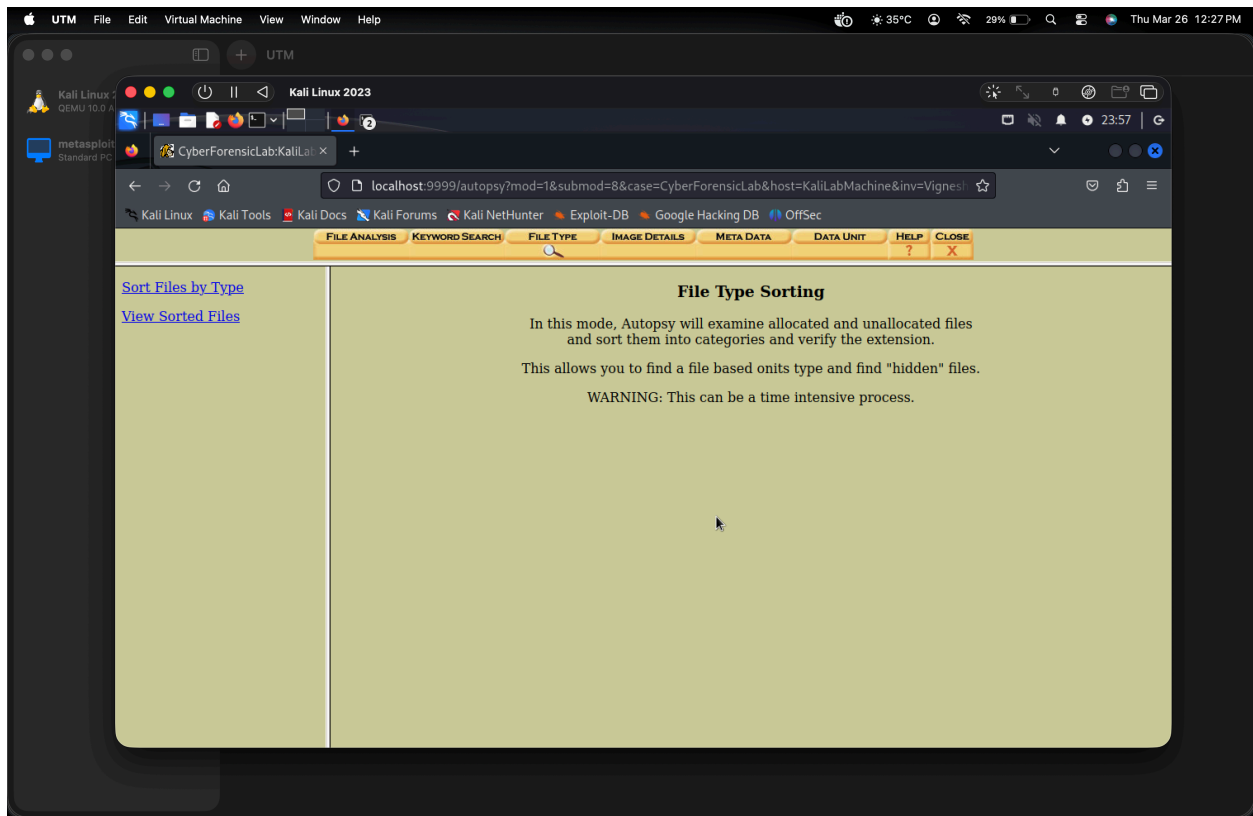












## Step 9: Analyze Evidence

